

Daksh Daksh

PhD Candidate, MPIP Mainz, Germany (daksh-daksh.github.io)

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EDUCATION

Max Planck Institute for Polymer Research, Mainz, Germany
PhD Computer Science Oct 2021 – present
Advisor: Prof. Katharina Landfester & Dr. Ingo Lieberwirth

National Forensic Sciences University (AIR – 159 in Computer Sciences),
Gandhinagar, India
MS - Forensic Nanotechnology (Nano-engineering) Jul 2014 – May 2016
Graduated in First class with Distinction

Lovely Professional University, Phagwara, India
B.Tech (Honors) Computer Science & Engineering Jul 2008 – May 2012
Graduated in First class with Distinction

Research Experience

PhD Researcher & affiliated with CRC 1066, MPIP Mainz 2021 – Present

- Working on automated image registration for correlative microscopy using deep learning.
- Developed Python tool for training free multi-modal image registration of light and electron microscopy and supervised image preprocessing tool for generating the precised image-pair.
- Worked on deep Learning for image analysis and microscopy data analysis using Machine Learning e.g. for generation of super resolution microscopy images

Web group Coordinator, Max Planck PhDnet, MPG 01/2023 – 01/2025

- Development and maintenance of the PhDnet website and mailing lists.

Research fellow in CSIE & IEO dept., NTNU, Taiwan 05/2018 – 07/2020

Jointly worked on project with NVIDIA AI Tech, Taiwan

- Worked on optical defect detection using supervised deep learning for Digital Holography.
- Worked on data Augmentation using GAN i.e. AC-GAN and DC-GAN and for Deep Learning Based on Digital Holograms.
- Worked on deep Learning for image analysis and generating different holographic image datasets.

Scientific Co-worker, University of Greifswald, Germany 06/2017 – 02/2018

- Worked on computational modeling and simulations for bio-molecules.

Setup configuration specialist, Aon Hewitt Pvt. Ltd., India (got selected among 5000 students & even before final semester) 11/2012 – 07/2014

- Worked on IBM mainframes, Lotus notes, C++, ASP .net (C#), VB, and SQL

Conferences	Multi-resolution Cross-modality Image Registration Using Unsupervised Deep Learning Approach. <u>D Daksh</u> , A Kaltbeitzel, K Landfester, I Lieberwirth Microscopy and Microanalysis: Microscopy Society of America 2023	
	Unsupervised Deep Learning approach for image registration in Correlative Microscopy for the localization of Nanoparticles. <u>D Daksh</u> , A Kaltbeitzel, G Glaßer, I Lieberwirth, K Landfester Invited Speaker, BIO Web of Conferences. European Microscopy Congress 2024	
	Correlative Microscopy Strategies for the Identification of Intracellular Nanoparticles and their Cellular Processing. I Lieberwirth, S Han, A Kaltbeitzel, G Glaßer, <u>D Daksh</u> , K Landfester Microscopy and Microanalysis: Microscopy Society of America 2024	
Patent	Multimodal Training-Free Registration Using Mutual Image Information. Application submitted. Software tool & licensing with Max Planck Innovations.	
Symposium & Summer schools	Microscopy data analysis: machine learning and the BioImage Archive. EMBL Heidelberg, Germany May 2022	
	First EMBL Imaging centre Symposium: Enabling imaging across scales. EMBL Heidelberg, Germany May 2022	
	Oral short talk	
	SFB 1066 and 1278 Joint Summer school. Fulda, Germany	July 2022
	Poster Presentation	
	Joint Symposium of RMaP and SFB 1066. Mainz, Germany	July 2022
	Poster Presentation	
	EMBO workshop: from Cryo-EM to multi-scale modelling. EMBL Heidelberg, Germany	February 2023
	Poster Presentation	
	Deep Learning & computer vision (DLCV) school. Genova, Italy June 2023 Selected among 1% with full scholarship to attend. Won the first prize for the most novel idea and best presentation.	
	Joint Symposium of SFB1066 and TRR319. Mainz, Germany	April 2024
	Poster Presentation	
	Internal Symposium of SFB 1066. Mainz, Germany	July 2024
	Poster Presentation	
	International Symposium of SFB 1066. Mainz, Germany	January 2025
	Poster Presentation	
Certificates	Optimizing writing strategies. MPIP Mainz, Germany	feb. 2024
	German B1. In-house course in MPIP Mainz, Germany	Sept. 2024
	Python for HPC. MPCDF Garching, Germany	Nov. 2024
	Parallel Computing With Matlab. MPCDF Garching, Germany	Dec. 2024
	Workshop on AI and Research in MPG. Berlin, Germany	Dec. 2024
Skills	Programming Languages & Tools: Python, PyTorch, Pandas, Anaconda navigator with different IDEs, Matlab, Napari, ImageJ, Bash, Git, LaTeX, TensorFlow, Scikit-learn, Scipy, Numpy	
	Languages: English (fluent), Hindi (native), German (B1), French (beginner)	